

求极限

$$\lim_{n \rightarrow \infty} n^5 \iint_D xy^2(1-x-y)^n dx dy$$

$$D = \{(x, y) | x \geq 0, y \geq 0, x + y \leq 1\}$$

换元, 令 $s = x + y, t = x - y$

得到

$$\begin{aligned} \iint_D xy^2(1-x-y)^n dx dy &= \frac{1}{16} \int_0^1 (1-s)^n \int_{-s}^s (s^2 - t^2)(s-t) dt ds \\ &= \frac{1}{12} \int_0^1 s^4 (1-s)^n ds \\ &= \frac{1}{12} B(5, n+1) = \frac{\Gamma(5)\Gamma(n+1)}{12\Gamma(n+6)} \\ &= \frac{10}{(n+5)(n+4)(n+3)(n+2)(n+1)} \end{aligned}$$

那么

$$\begin{aligned} \lim_{n \rightarrow \infty} n^5 \iint_D xy^2(1-x-y)^n dx dy \\ = \frac{10n^5}{(n+5)(n+4)(n+3)(n+2)(n+1)} = 10 \end{aligned}$$